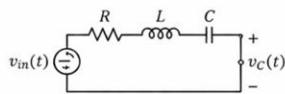


## Series RLC Circuit — Why Laplace Transform (LT)?

One circuit, four views: time-domain → LT (s-domain) → simplified solution



**Series RLC circuit (output = capacitor voltage  $v_C(t)$ )**  
Input  $v_{in}(t)$  excites a resistor  $R$ , inductor  $L$ , capacitor  $C$  in series.  
**Goal:** Find  $v_C(t)$  for a given input  $v_{in}(t)$  (with zero initial conditions)

### 1) Time-domain model (Differential Equation) (from KVL + element laws)

$$L \frac{d^2 v_C(t)}{dt^2} + R \frac{dv_C(t)}{dt} + \frac{1}{C} v_C(t) = \frac{dv_{in}(t)}{dt}$$

This is a linear differential equation with constant coefficients.

**Direct time-domain solution? → Hard** (especially for arbitrary  $v_{in}(t)$ )

You'd use convolution integrals or guess-and-check.

**This is where LT helps**

### 2) Take Laplace Transform (zero initial conditions)

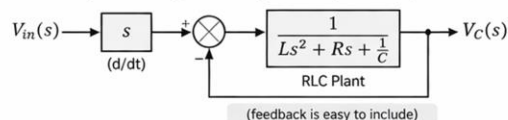
Derivatives → algebraic terms,  $\mathcal{L}\{f(t)\} = sF(s)$

$$\left(Ls^2 + Rs + \frac{1}{C}\right) V_C(s) = s V_{in}(s)$$

**Algebraic equation in s-domain**

Much easier to manipulate (just algebra).

### 3) Block-Diagram Manipulation (s-domain)



(feedback is easy to include)

In s-domain, derivatives become multiplications by  $s$  → Simple interconnection

### 4) Transfer Function (Output/Input)

$$\frac{V_C(s)}{V_{in}(s)} = \frac{s}{Ls^2 + Rs + \frac{1}{C}} \quad \text{Done. Now analyze in seconds.}$$

→ Poles: roots of  $Ls^2 + Rs + \frac{1}{C} = 0$

→ **Stability:** stable if  $R > 0, L > 0, C > 0$  (all poles in LHP)

→ Step response:  $V_{in}(s) = \frac{V_0}{s} \rightarrow V_C(s) = \frac{V_0}{Ls^2 + Rs + \frac{1}{C}} \rightarrow$  inverse LT (table/partial fractions)

### Why use LT?

- Turns differential equations into **algebraic equations**
- Makes **block-diagram methods possible** (easy feedback, cascades, etc.)
- Direct path to **transfer function, poles, stability**, and standard responses

LT doesn't change the system.  
**It simplifies the analysis.**



## Robotics Systems: Circuit → DE → Laplace → Block Diagram

Why we use the Laplace Transform (LT) → It turns calculus (d/dt, integrals) into algebra (s)

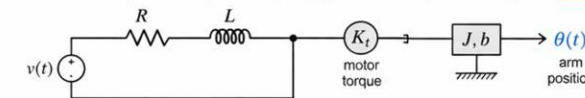
Same physics, easier math, clearer design.



### 1) Physical system → Circuit model

A DC motor drives a joint via a gear. Voltage  $v(t)$  produces arm position  $\theta(t)$

**Standard modeling step**



•  $R$ : armature resistance •  $L$ : armature inductance •  $K_t$ : torque constant •  $J$ : inertia •  $b$ : viscous friction

### 2) From circuit → Differential equation (time domain)

Apply KVL (electrical) + torque balance (mechanical):

$$\underbrace{L \frac{di(t)}{dt}}_{\text{inductor voltage}} + \underbrace{R i(t)}_{\text{resistor voltage}} + \underbrace{K_t \omega(t)}_{\text{motor back-EMF}} = v(t) \quad \Rightarrow \quad \underbrace{J \frac{d\omega(t)}{dt}}_{\text{inertia torque}} + \underbrace{b \omega(t)}_{\text{friction torque}} = K_t i(t) \quad \Rightarrow \quad \theta(t) \quad (\text{position})$$

**This is a coupled linear differential equation (DE) in time.**

We could solve it—but DEs get messy fast (especially with controllers, gears, sensors).

### 3) Why use Laplace Transform (LT)? → Algebra instead of calculus

Take the Laplace Transform of both sides (zero initial conditions for clarity):

**Time domain (differential equations)**

$$L \frac{di(t)}{dt} + R i(t) + K_t \omega(t) = v(t)$$

$$J \frac{d\omega(t)}{dt} + b \omega(t) = K_t i(t)$$

Derivatives → calculus

**Laplace domain (algebraic equations)**

$$(Ls + R) I(s) + K_t \Omega(s) = V(s)$$

$$(Js + b) \Omega(s) = K_t I(s)$$

Derivatives → **multiply by s** **Algebra!**

**Why this helps:**

- ✓ No derivatives to carry around
- ✓ Coupled DEs → simple linear algebra
- ✓ Easy to include controllers

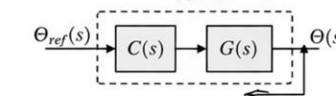
### 4) Block diagram (control view)

In s-domain, we express the motor + arm as a transfer function:

$$V(s) \rightarrow \frac{K_t}{(Ls + R)(Js + b) + K_t^2} \rightarrow \Theta(s)$$

$$\text{Plant: } G(s) = \frac{\Theta(s)}{V(s)}$$

**Add a controller  $C(s)$  (e.g., PID)**



**Closed-loop transfer:**

$$\frac{\Theta(s)}{\Theta_{ref}(s)} = \frac{C(s)G(s)}{1 + C(s)G(s)} \quad \text{Just algebra.}$$

Try doing this directly in time domain → **much harder.**

### Takeaway

Laplace Transform converts differential-equation robot models into **algebraic transfer functions and block diagrams.**

That's why it's the standard tool for robotics analysis & control design.